

Near-Light Photometric Stereo for Endoscopic Bone Surface Measurement: Ring-LED Calibration, Offset-Scale Geometry, and Fiducial-Anchored Registration to CT

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Abstract—Background—Image-guided intervention on bone requires a spatial correspondence between the intra-operative endoscopic view and a pre-operative CT volume. Establishing that correspondence demands metric three-dimensional surface information, which a monocular endoscope does not natively provide. Photometric stereo is attractive because the emitters are already integrated into the instrument tip, but its classical formulation assumes distant, directional illumination — an assumption that endoscopic geometry violates severely. **Purpose**—To present a near-light photometric stereo pipeline for a four-LED ring endoscope, comprising the ring geometry and its offset and scale calibration, a calibrated point-source forward model, a perspective log-depth solver, retroreflective fiducial localization, and fiducial-anchored transfer into the CT frame; and to quantify, against CT ground truth, the error incurred by the far-field approximation at this geometry. **Methods**—Four LEDs are mounted 90° apart on a ring of measured radius 12.08 mm about the lens and sequenced individually by a microcontroller, giving four single-emitter exposures plus an ambient frame per session at a measured 30 mm working distance over a 37 mm field. The offset-to-working-distance ratio is 0.40, placing the rig unambiguously in the near field. We derive the perspective log-depth relation used to integrate normals metrically, and implement the LED calibration of Quéau *et al.* recovering per-emitter position, principal direction, anisotropy and intensity. Five sessions of an ex-vivo vertebral specimen were processed; one carries CT ground truth for three fiducials, transferred into the camera frame by a P3P solution disambiguated with a confirmed ordinal constraint. **Results**—Fiducial localization recovered 18 markers, 14 (78%) confirmed independently under all four illumination directions. Against CT, the far-field depth estimate proved *anti-correlated* with true lens-to-marker distance ($r = -0.923$), reversing the near-to-far ordering of all three CT-known fiducials; least-squares affine rescaling onto the CT control points returned a *negative* slope and left a 4.36 mm RMS residual over a 20.7 mm true span. The near-light solver and its calibration pass synthetic

self-tests to 0.00 mm in emitter position and 0.090° in principal direction. **Conclusions**—Fiducial localization and CT/P3P distances are reliable and serve as registration inputs. The far-field depth channel is not merely imprecise but sign-inverted at this geometry, and must not propagate into a registration result; the calibrated near-light formulation is the corrective path. [REGISTRATION RESULT PENDING]

Index Terms—Photometric stereo, near-field illumination, endoscopy, LED calibration, retroreflective fiducials, image registration, surgical navigation.

I. INTRODUCTION

REGISTERING an endoscopic view to a pre-operative CT volume is the enabling step for image-guided intervention on bone. The problem is well posed only when the endoscopic side of the correspondence carries three-dimensional, metrically-scaled information. A monocular endoscope does not supply this directly, so it must be recovered — from motion, from structured illumination, or, as here, from shading.

Photometric stereo recovers surface orientation from the way brightness varies as the illumination direction changes. It suits endoscopy because the light sources are already present at the instrument tip: no additional hardware enters the surgical field, and no relative motion between camera and specimen is required. The classical formulation, however, assumes *far-field* illumination — that each source is distant enough for its direction to be constant across the field of view and for its intensity not to fall off within the scene.

Endoscopic rigs violate both assumptions comprehensively. On the instrument described here the emitters sit 12.08 mm off the optical axis at a 30 mm working distance, a ratio of 0.40. The direction to a given emitter therefore sweeps by tens of degrees across the 37 mm field, and inverse-square falloff is a first-order effect rather than a correction term. The correct treatment is *near-light* photometric stereo: a calibrated point-source forward model with per-pixel light vectors, solved under perspective rather than orthographic projection.

This paper presents that pipeline end to end, and reports as its principal empirical contribution a CT-validated measurement of what the far-field shortcut costs at this geometry. The answer is worse than a scale error: the

Manuscript draft v0.3. Sections on endoscope-to-CT registration (Sec. VII, Sec. VIII-G) are marked placeholders and are not yet complete. All other reported values are reproduced from the committed pipeline outputs and are traceable to a named file and column. Every derivation in Sec. III–V was independently re-derived and numerically cross-checked, and the caveats reported throughout — including the scale conflict of Sec. VIII-E and the self-test limitations of Sec. VIII-F — are outcomes of that audit.

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recovered depth is *anti-correlated* with truth, reversing the depth ordering of the fiducials outright. We regard this as the more useful contribution, because the far-field shortcut is widely taken in endoscopic shading work and the *sign* of the resulting error appears not to have been measured against ground truth in this configuration.

A. Contributions

- 1) A complete near-light photometric stereo pipeline for a ring-LED endoscope, with the ring geometry, its offset and scale calibration, and the resulting camera model stated as measured quantities with explicit provenance (Sec. II–III).
- 2) A self-contained derivation of the perspective log-depth integration used to recover metric depth from calibrated near-light normals (Sec. IV-F).
- 3) An implementation and synthetic validation of per-emitter photometric calibration — position, principal direction, anisotropy, intensity — following Quéau *et al.* [1] (Sec. V).
- 4) A retroreflective fiducial detector exploiting the glass-bead sparkle signature, with a cross-frame confirmation statistic (Sec. VI).
- 5) A quantified, CT-validated demonstration that far-field photometric stereo inverts depth ordering at endoscopic near-light geometry (Sec. VIII-C).
- 6) **[REGISTRATION CONTRIBUTION — PENDING, Sec. VII]**

II. SYSTEM DESIGN

A. Optical head and LED ring

The instrument is a monocular endoscope with four LEDs mounted on a ring concentric with the lens, spaced 90° apart in azimuth (Fig. 1a). Writing L_i for the emitter at azimuth α_i , the layout is

$$\alpha_1 = 0^\circ, \quad \alpha_2 = 90^\circ, \quad \alpha_3 = 180^\circ, \quad \alpha_4 = 270^\circ, \quad (1)$$

all at the same measured ring radius r_{LED} . In the camera frame (Sec. II-C) the emitter centres lie in the lens plane $Z = 0$ at

$$\mathbf{x}_s^{(i)} = (r_{\text{LED}} \cos \alpha_i, -r_{\text{LED}} \sin \alpha_i, 0)^\top, \quad (2)$$

the minus sign on the second component arising because azimuth is measured counter-clockwise from $+X$ while Y points *down*.

The four emitters are nominally identical. Equal emitter power is not assumed by the near-light solver, which takes per-emitter intensity from calibration (Sec. V); where emitters are genuinely identical the common factor folds into albedo and cancels in the normalization $\mathbf{n} = \mathbf{g}/\|\mathbf{g}\|$.

B. Illumination sequencing

The ring is driven by an Arduino Pro microcontroller, which switches exactly one emitter at a time and holds the remaining three dark, so that each exposure isolates a single known illumination direction. A fifth exposure is captured

TABLE I
ACQUISITION INVARIANTS WITHIN ONE CAPTURE SESSION.

Quantity	Status	Consequence
Camera pose	held constant	frames pixel-registered by construction
Specimen pose	held constant	a fiducial occupies the same pixel in every frame
Illumination direction	varied (4 emitters)	the photometric signal
Ambient light Exposure	measured once nominally fixed, normalized in software	subtracted per frame Sec. IV-D

with all emitters off to record the ambient component. Because the sequence is electronically controlled while the camera and specimen remain stationary, the five frames of a session are pixel-registered by construction — a property the fiducial consolidation of Sec. VI-C exploits directly.

Hardware details to record

The specific Arduino Pro variant, the LED part number, the drive current, and the exposure/frame timing are not yet recorded in the repository. The LED datasheet half-power angle $\theta_{1/2}$ is required by the anisotropy step of Sec. V-A and must be obtained from the part datasheet. See Appendix A, item C1.

C. Coordinate conventions

All quantities use a single right-handed camera frame: X to the right, Y downward, Z from the lens into the scene, origin at the optical centre, units of millimetres. Azimuth is measured counter-clockwise from $+X$ in the lens plane. Depth Z is positive into the scene, so a point farther from the lens has larger Z . Outward-facing surface normals — those pointing back toward the camera — therefore have $n_3 < 0$, a convention used throughout Sec. IV and enforced explicitly in the solver.

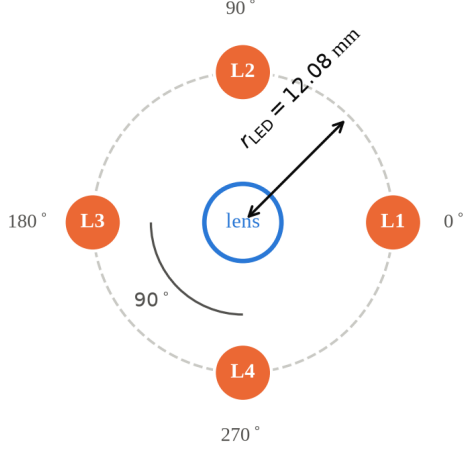
D. Capture protocol

Each session ("shot") is a directory of five exposures of a static scene: `led1.png–led4.png`, one per emitter, and `dark.png` with all emitters off. A session is admitted to processing only if all four single-emitter frames are present. Table I states the acquisition contract explicitly, since every downstream stage depends on it.

III. GEOMETRIC CALIBRATION: OFFSET AND SCALE

Two measured lengths govern the entire pipeline: the ring *offset* r_{LED} , which sets the illumination geometry and the near-field regime; and the field *scale*, which converts pixels to millimetres and fixes the camera's focal length. Table II lists every geometric parameter with its provenance — a distinction that matters here because the measured and calibration-derived values disagree (Sec. III-C).

(a) LED ring, face view



(b) side view (Z into the scene)

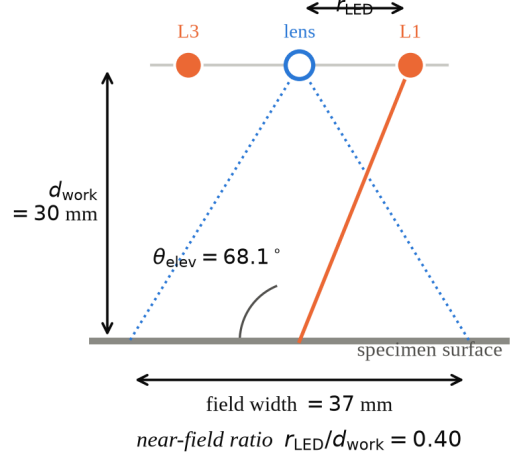


Fig. 1. System geometry. (a) Face view of the LED ring: four emitters 90° apart at the measured radius $r_{\text{LED}} = 12.08 \text{ mm}$ about the lens. (b) Side view showing the measured working distance $d_{\text{work}} = 30 \text{ mm}$, the 37 mm field width, and the resulting illumination elevation $\theta_{\text{elev}} = 68.1^\circ$ at the on-axis surface point. The offset-to-working-distance ratio of 0.40 is what places this rig in the near field and makes the far-field approximation inadmissible (Sec. IV-A). All dimensions are measured values.

TABLE II
RIG AND IMAGING PARAMETERS. FRAME: X RIGHT, Y DOWN, Z INTO THE SCENE, ORIGIN AT THE OPTICAL CENTRE.

Symbol	Value	Provenance	Role
r_{LED}	12.08 mm	measured	ring offset
field width	37.0 mm	measured	scale
d_{work}	30.0 mm	measured	working distance
θ_{elev}	68.1°	derived	elevation
s	0.05781 mm/px	derived	in-plane scale
f_x	519 px	derived	focal length
f_x (chkb.)	794 px	calibrated	conflicts (Sec. III-C)
W	640 px	config.	working resolution
$\theta_{1/2}$	[TBD]	datasheet	emitter anisotropy

A. The offset and the illumination elevation

The elevation of an emitter above the surface plane, evaluated at the surface point on the optical axis, follows from the right triangle formed by the ring offset and the working distance:

$$\theta_{\text{elev}} = \arctan\left(\frac{d_{\text{work}}}{r_{\text{LED}}}\right) = \arctan\left(\frac{30.0}{12.08}\right) = 68.1^\circ. \quad (3)$$

This value is *derived* from the two measured lengths rather than assumed. It is, however, the *on-axis* elevation, and it is the single constant the far-field baseline applies to every pixel. Across the 37 mm field along the emitter's axis the true elevation ranges from 44.5° through 90° to 77.9° ; evaluated at the three CT-known fiducial positions in the recovered camera frame it takes the values 74.9° ,

83.5° and 57.9° , against the constant 68.1° actually used. That spread is the far-field approximation made visible, and it is precisely the point of Sec. IV-A.

B. The scale and the camera model

Images are undistorted and resized so the long edge is $W = 640 \text{ px}$; a pinhole model with

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

is used thereafter, with the principal point at the image centre. The in-plane scale at the working distance is

$$s = \frac{\text{field width}}{W} = \frac{37.0}{640} = 0.05781 \text{ mm/pixel}. \quad (5)$$

Rather than adopt the checkerboard focal length, the pipeline derives f_x from the same two measured lengths. For a pinhole camera imaging a field of metric width Ω at distance d onto W pixels, similar triangles give $\Omega/d = W/f_x$, hence

$$f_x = \frac{W d_{\text{work}}}{\Omega} = \frac{640 \times 30.0}{37.0} = 519 \text{ pixel}. \quad (6)$$

The vertical focal length follows from the same distance and the field height $\frac{3}{4}\Omega$, which encodes a 4:3-shaped physical field irrespective of the frame's aspect ratio; at the 4:3 working resolution actually used (640×480) it correctly yields $f_y = f_x$. Note that Eqs. (5) and (6) are consistent by construction: $d_{\text{work}}/f_x = F/W = s$ identically.

Two caveats on the scale must be stated. First, s is the scale *at the working distance only*: the true in-plane

scale is Z/f_x and therefore varies with depth, so applying the constant s across a depth map is itself a small-relief approximation. Second, the principal point is set to the image centre, whereas the calibration — transported to the undistorted working frame — places it near (324.1, 225.5), a vertical discrepancy of 14.5 pixel. The docstring’s stated intent is that only the absolute focal length be replaced; in fact c_x, c_y are silently replaced too. The effect on the reported CT distances is 0.1–0.7 mm — small, but a straightforward correctness fix.

C. The focal-length ambiguity

The checkerboard calibration returns $f_x = 1588$ pixel at native resolution, i.e. 794 pixel at the 640 px working width — a factor of 1.53 above Eq. (6). Equivalently, the calibrated focal length implies a 45.9 mm working distance for a 37 mm field where 30 mm was measured, and a field of view near 44° where the measured pair imply roughly 63° .

We resolve this in favour of the bench measurements, on the grounds that the two measurements are mutually consistent while the checkerboard result contradicts both. This is an arbitration, not a resolution. Every *absolute* distance in this paper therefore carries an unresolved $\sim 1.5\times$ scale uncertainty; relative comparisons within a session are unaffected, since the ambiguity is a single multiplicative factor.

Calibration residuals not recorded

The checkerboard reprojection error, the number and spread of calibration poses, and the distortion coefficients are not currently recorded. A calibration reported without its residuals cannot be assessed, and the residual may itself explain the $1.53\times$ discrepancy. The ambiguity is settled definitively by imaging an object of known size at a known distance. See Appendix A, item C2.

IV. NEAR-LIGHT PHOTOMETRIC STEREO

A. Why the far-field model is inadmissible here

Far-field photometric stereo requires that the direction to each source be constant over the field and that intensity falloff within the scene be negligible. The governing dimensionless quantity is the ratio of source offset to working distance,

$$\eta = \frac{r_{\text{LED}}}{d_{\text{work}}} = \frac{12.08}{30.0} = 0.40, \quad (7)$$

which is order unity rather than small. Two consequences follow directly. First, the direction from a surface point to a given emitter varies across the 37 mm field by tens of degrees, so no single constant light vector describes the emitter. Second, the emitter-to-surface distance varies by well over 10% across the same span, so the $1/r^2$ term cannot be absorbed into a constant.

Neglecting both produces a smooth, low-spatial-frequency brightness bias that the normal-recovery step

misinterprets as surface tilt. Integration then accumulates that spurious tilt into a large-scale curvature which dominates the genuine relief. Section VIII-C measures the result against CT: at this geometry the error is not a scale factor but a sign inversion.

B. Near-light irradiance model

We adopt the LED model of Quéau *et al.* [1]. For emitter i with position $\mathbf{x}_s^{(i)}$, unit principal direction $\mathbf{n}_s^{(i)}$, anisotropy exponent μ_i and intensity Ψ_i , the irradiance at a surface point $\mathbf{x}$ with unit normal $\mathbf{n}$ and albedo ρ is

$$I^{(i)} = \Psi_i \rho \underbrace{\left[\frac{\mathbf{n}_s^{(i)} \cdot (\mathbf{x} - \mathbf{x}_s^{(i)})}{\|\mathbf{x} - \mathbf{x}_s^{(i)}\|} \right]^{\mu_i}}_{\cos^{\mu_i} \theta, \text{ angular profile}} \underbrace{\frac{\{(\mathbf{x}_s^{(i)} - \mathbf{x}) \cdot \mathbf{n}\}_+}{\|\mathbf{x}_s^{(i)} - \mathbf{x}\|^3}}_{\text{Lambert} \times 1/r^2}, \quad (8)$$

where $\{\cdot\}_+$ denotes clamping at zero (self-shadowing) and θ is the emission angle off the emitter’s principal axis. The cubic power in the denominator is not a typographical accident: one power normalizes the un-normalized dot product in the numerator into a cosine, leaving $1/r^2$ attenuation.

It is convenient to gather the geometry into a per-pixel *effective light vector*. Writing $\mathbf{d} = \mathbf{x}_s - \mathbf{x}$, $r = \|\mathbf{d}\|$ and $\boldsymbol{\omega} = \mathbf{d}/r$, Eq. (8) becomes $I^{(i)} = \rho \mathbf{v}^{(i)} \cdot \mathbf{n}$ with

$$\mathbf{v}^{(i)}(\mathbf{x}) = \Psi_i \frac{(\max(-\boldsymbol{\omega} \cdot \mathbf{n}_s^{(i)}, 0))^{\mu_i}}{r^2} \boldsymbol{\omega}. \quad (9)$$

Every factor in Eq. (9) — $\mathbf{x}_s$, $\mathbf{n}_s$, μ , Ψ — comes from calibration (Sec. V); none is assumed from rig symmetry. This is the essential difference from the far-field baseline, in which $\mathbf{v}^{(i)}$ is a single constant unit vector for the whole image.

C. Specular-free channel

Bone is wet and glossy, and the retroreflective fiducials are strongly non-Lambertian, so the Lambertian model of Eq. (8) cannot be applied to raw intensity. We use a dichromatic separation: under the dichromatic reflection model the surface-reflection (specular) component carries the illuminant’s spectrum while the body-reflection component carries the surface spectrum. For an illuminant whose linear red and blue responses are equal, the difference

$$I_{\text{body}} = \max(I_R^{\text{lin}} - I_B^{\text{lin}}, 0) \quad (10)$$

cancels the specular term and retains a signal proportional to body reflection. Writing $I_c = m_d B_c + m_s E_c$ for channel c , with m_d, m_s the geometric body and surface factors and B_c, E_c the corresponding spectral integrals,

$$I_R - I_B = m_d(B_R - B_B) + m_s(E_R - E_B), \quad (11)$$

so the specular term vanishes *iff* $E_R = E_B$. The residual $(B_R - B_B)$ is a per-pixel material constant identical across the four frames, so it folds into the pseudo-albedo and cancels in $\mathbf{n} = \mathbf{g}/\|\mathbf{g}\|$.

The precise assumption, and its sensitivity

The requirement is not that the emitter be “white” colorimetrically, but that it produce *equal linear R and B responses in this camera*, after whatever white balance and colour matrix the pipeline applies. Phosphor-converted white LEDs have a strongly non-flat spectrum, so this holds by construction only if the camera is balanced on a neutral target *under these emitters*. The failure is not graceful: a 5% R/B imbalance leaks $\sim 18\%$ of the specular signal into the body channel, and that leak has exactly the specular lobe’s geometry — concentrated where it does most damage. The measured E_R/E_B of the rig on a neutral target should be reported.

A second, smaller effect: clamping the difference at zero is a nonlinear operation that rectifies noise where the signal is near zero, i.e. in shadow and background. In simulation this biases shadowed pixels $\sim 25\%$ high while leaving well-lit pixels unbiased — and shadowed pixels are exactly where a K -light solve is already most fragile.

D. Exposure normalization

Camera gain is a single global multiplier that scales all channels equally, so it is estimated from linear luminance — which stays positive in shadow — over the well-lit region, and the body-reflection frame is divided by that scalar.

This detail is worth reporting because the obvious alternative is wrong. Normalizing each frame by the median of its *own* specular-free channel fails: that channel is near zero over dark background and single-emitter shadow, so its median can collapse toward zero, and dividing by it injects a large spurious per-emitter gain. A factor of $\sim 250\times$ was observed from this mechanism before the estimator was moved onto luminance.

E. Per-pixel normal recovery

For K emitters the model is linear in the albedo-scaled normal $\mathbf{g} = \rho \mathbf{n}$. Stacking the per-pixel effective light vectors of Eq. (9) as rows of $M \in \mathbb{R}^{K \times 3}$ and the observed body intensities as $\mathbf{I} \in \mathbb{R}^K$,

$$M\mathbf{g} = \mathbf{I}, \quad \mathbf{g} = (M^\top M + \epsilon \mathbb{I})^{-1} M^\top \mathbf{I}, \quad (12)$$

with a small ϵ for conditioning. The normal equations are used so the same code path serves $K = 3$ (exact) and $K = 4$ (least squares, the ring rig here) without special-casing.

Two properties of this system deserve statement rather than assumption. The emitters sit only 21.9° off the optical axis at 12.08 mm and 30 mm, so the configuration is inherently narrow-baseline: $\text{cond}(M) \approx 5.4$ on axis, and forming the normal equations squares it to $\text{cond}(M^\top M) \approx 28.8$, rising to 49.7 off axis. And the regularizer is *absolute*: at millimetre scale $\|\mathbf{v}\| \sim \Psi/r^2$ is small enough that a fixed $\epsilon = 10^{-9}$ amounts to 0.5–2.9% of the smallest eigenvalue, inducing a systematic, spatially varying normal tilt of 0.04 – 0.30° that grows with depth and off-axis distance. A relative

form $\epsilon = \lambda \text{tr}(M^\top M)/3$ would be scale-invariant; as written the term is physically scaled, not merely numerical, and would silently change meaning if the code were switched to metres. Albedo and unit normal follow as

$$\rho = \|\mathbf{g}\|, \quad \mathbf{n} = \mathbf{g}/\rho, \quad (13)$$

and the sign is fixed by the outward convention of Sec. II-C: any normal with $n_3 > 0$ is negated, so that all normals face the camera.

Crucially, M varies from pixel to pixel because $\mathbf{v}^{(i)}$ depends on the surface point $\mathbf{x}$ — which is what the solver is trying to find. This circularity is resolved by iteration (Sec. IV-G).

F. Perspective log-depth integration

Given normals, depth must be recovered by integration. Under *orthographic* projection one integrates $\partial z/\partial u = n_1/n_3$ directly; that is systematically wrong here, because the relief is a substantial fraction of the working distance and the metric footprint of a pixel therefore changes across the surface. The correct treatment is perspective.

Let $x_t = (u - c_x)/f_x$ and $y_t = (v - c_y)/f_y$, so that a surface point imaged at pixel (u, v) with depth $Z(u, v)$ is

$$\mathbf{P}(u, v) = Z(u, v) (x_t, y_t, 1)^\top. \quad (14)$$

Differentiating along the image axes, and using $\partial x_t/\partial u = 1/f_x$ and $\partial y_t/\partial u = 0$,

$$\frac{\partial \mathbf{P}}{\partial u} = \frac{\partial Z}{\partial u} (x_t, y_t, 1)^\top + Z \left(\frac{1}{f_x}, 0, 0 \right)^\top, \quad (15)$$

$$\frac{\partial \mathbf{P}}{\partial v} = \frac{\partial Z}{\partial v} (x_t, y_t, 1)^\top + Z \left(0, \frac{1}{f_y}, 0 \right)^\top. \quad (16)$$

The normal is orthogonal to both surface tangents, $\mathbf{n} \cdot \partial_u \mathbf{P} = \mathbf{n} \cdot \partial_v \mathbf{P} = 0$. Substituting and writing

$$D \equiv \mathbf{n} \cdot (x_t, y_t, 1)^\top = n_1 x_t + n_2 y_t + n_3, \quad (17)$$

the first condition gives $D \partial_u Z + Z n_1/f_x = 0$, hence

$$\frac{1}{Z} \frac{\partial Z}{\partial u} = -\frac{n_1}{f_x D}, \quad \frac{1}{Z} \frac{\partial Z}{\partial v} = -\frac{n_2}{f_y D}, \quad (18)$$

and since $\partial(\ln Z)/\partial u = Z^{-1} \partial Z/\partial u$,

$$\boxed{\frac{\partial \ln Z}{\partial u} = -\frac{n_1}{f_x D}, \quad \frac{\partial \ln Z}{\partial v} = -\frac{n_2}{f_y D}.} \quad (19)$$

The problem is therefore linear in $\ln Z$, not in Z . The two gradient fields of Eq. (19) are integrated by an FFT Poisson solve enforcing integrability, and the result is exponentiated.

Two implementation points matter. First, D is clamped away from zero and held negative ($D \leq -0.10$), consistent with the outward convention $n_3 < 0$; this bounds the gradient where the surface turns away from the camera. Second, integration determines $\ln Z$ only up to an additive constant, so Z is determined only up to one global multiplicative scale. That scale is fixed by anchoring the

median depth over the specimen mask to the measured working distance:

$$Z = d_{\text{work}} \exp(\ln Z' - \text{median}_{\mathcal{M}} \ln Z'), \quad (20)$$

where $\ln Z'$ is the integrated field and $\mathcal{M}$ the specimen mask. Because $\exp$ is strictly monotonic it maps the middle order statistic to the middle order statistic, so Eq. (20) sets the median of Z over $\mathcal{M}$ to exactly d_{work} for an odd number of mask pixels. For an even count the usual tie-break averages the two central values, and $\exp$ of their arithmetic mean is their geometric mean, differing by a factor $\cosh((b-a)/2)$; for adjacent order statistics of $\sim 10^5$ pixels this is a relative discrepancy below 10^{-12} and is negligible.

Two honest qualifications belong with this step. First, integration recovers $\ln Z$ exactly only when (p, q) is curl-free; with noisy normals it is not, and the FFT solver returns the least-squares integrable approximation rather than the field itself. Second, because the scale is re-anchored to d_{work} on every sweep, the near-light model buys correct *perspective foreshortening*, not absolute depth — per-pixel albedo is unknown, so albedo and r trade off and absolute scale is imposed rather than solved.

G. Fixed-point iteration

Because the light vectors depend on the unknown geometry, the solver alternates:

- 1) initialize Z to the constant working distance;
- 2) back-project to points $\mathbf{P} = Z(x_t, y_t, 1)^\top$;
- 3) evaluate per-pixel light vectors, Eq. (9);
- 4) solve for normals, Eq. (12);
- 5) integrate to depth, Eqs. (19)–(20);
- 6) repeat from step 2.

Two iterations are used, and the choice is justified by a convergence sweep on the synthetic scene rather than asserted. The alternation reaches a stable fixed point within three to four sweeps: median normal error and recovered relief are unchanged to four significant figures from sweep 4 through sweep 30, at two noise levels. The first sweep, which evaluates the light vectors on a flat plane at the working distance, systematically *over-estimates* relief — 114.9% of truth — because the $1/r^2$ falloff is evaluated at the wrong depth. One refinement corrects this to 99.6% (Table III). Two sweeps therefore sit within 0.11° median normal error and 0.8% relief of the converged solution at half the cost.

Wording to correct in the implementation

The solver’s docstring describes the loop as alternating “until stable” while the code in fact runs a fixed count with no stability test, and attributes the choice of two to a degradation at higher counts. The sweep above does not reproduce any such degradation. Either add an explicit convergence criterion or describe the constant as the cost/accuracy tradeoff it is; the current comment would not survive a reviewer re-running the self-test.

TABLE III

CONVERGENCE OF THE GEOMETRY-NORMALS ALTERNATION ON THE SYNTHETIC SCENE. THE SOLUTION IS A STABLE FIXED POINT FROM SWEEP 4 ONWARD, NOT A DEGRADING SEQUENCE.

Sweeps	median normal err.	relief ratio	median depth err.
1	2.628°	1.1486	0.268 mm
2	3.966°	0.9963	0.308 mm
3	3.855°	1.0044	0.306 mm
4	3.853°	1.0044	0.306 mm
10	3.854°	1.0044	0.306 mm
30	3.854°	1.0044	0.306 mm

V. PHOTOMETRIC CALIBRATION OF THE LED RING

The forward model of Eq. (8) introduces four parameters per emitter. Each has its own calibration procedure; none is assumed from ring symmetry.

A. Anisotropy from the datasheet

The imperfect-Lambertian emitter profile is $\Phi(\theta) = \Phi_0 \cos^\mu \theta$. The datasheet half-power angle $\theta_{1/2}$ is where relative intensity falls to one half, so $\cos^\mu \theta_{1/2} = \frac{1}{2}$ and

$$\mu = \frac{-\ln 2}{\ln \cos \theta_{1/2}}. \quad (21)$$

A source with $\theta_{1/2} = 60^\circ$ gives $\mu = 1$ exactly — a truly Lambertian emitter — and $\theta_{1/2} = 45^\circ$ gives $\mu = 2$; a narrower beam gives larger μ , with $d\mu/d\theta_{1/2} < 0$ throughout $(0, 90^\circ)$.

Datasheet convention — a live trap

Eq. (21) takes the *half*-angle $\theta_{1/2}$, but LED datasheets almost universally quote the *full* included viewing angle $2\theta_{1/2}$: a part advertised as “120° viewing angle” has $\theta_{1/2} = 60^\circ$ and hence $\mu = 1$. Feeding the datasheet number directly into Eq. (21) silently yields $\mu = 4.82$ instead of 2.00 for a 60°-quoted part. The convention must be stated explicitly alongside the recorded value.

B. Emitter position by mirror-sphere triangulation

A specular sphere of known radius R is imaged in several poses. For each pose, the sphere centre is recovered from its silhouette: the camera rays grazing the sphere form a right circular cone about the direction to the centre, so the boundary ray directions lie on a common plane $\mathbf{a} \cdot \mathbf{d} = \cos \beta$. The axis $\mathbf{a}$ is estimated as the least-variance eigenvector of the boundary directions, and the tangent-line geometry

$$\sin \beta = \frac{R}{\|\mathbf{C}\|} \implies \mathbf{C} = \frac{R}{\sin \beta} \mathbf{a} \quad (22)$$

fixes the range.

It is worth stating that $\cos \beta$ is estimated *exactly*, not approximately, and that the obvious alternative fails badly. Every boundary direction satisfies $\mathbf{a} \cdot \mathbf{d}_k = \cos \beta$ individually, so by linearity $\mathbf{a} \cdot \bar{\mathbf{d}} = \frac{1}{K} \sum_k \mathbf{a} \cdot \mathbf{d}_k = \cos \beta$ for *any* sampling of the boundary — uniform, clustered, or a short arc. The mean direction $\bar{\mathbf{d}}$ is however *not* a unit

vector: for uniform sampling $\hat{\mathbf{d}} = \cos \beta \mathbf{a}$, so normalizing it first and taking $\arccos(\mathbf{a} \cdot \hat{\mathbf{d}})$ returns $\beta = 0$ identically. The unnormalized projection is the correct estimator.

The specular highlight then gives a ray toward the emitter: the camera ray is traced to the near sphere intersection $\mathbf{s}$, the outward normal $\mathbf{n} = (\mathbf{s} - \mathbf{C})/R$ is formed, and the direction to the camera is reflected about it,

$$\ell = 2(\mathbf{n} \cdot \hat{\mathbf{c}}) \mathbf{n} - \hat{\mathbf{c}}, \quad \hat{\mathbf{c}} = -\mathbf{s}/\|\mathbf{s}\|, \quad (23)$$

so the emitter lies on the ray $\{\mathbf{s} + t\ell\}$. Across poses these rays do not meet exactly, so the position is the least-squares closest point, minimizing the squared perpendicular distance to each line:

$$\mathbf{x}_s = \arg \min_{\mathbf{x}} \sum_k \|(\mathbb{I} - \ell_k \ell_k^\top)(\mathbf{x} - \mathbf{s}_k)\|^2, \quad (24)$$

where $(\mathbb{I} - \ell \ell^\top)$ is the orthogonal projector onto the plane perpendicular to ℓ .

Conditioning must be reported, and currently is not

Both steps are exactly solvable but can be badly conditioned, and neither currently emits a diagnostic. *Silhouette coverage* dominates the position error: with the same detection noise and point count, a full silhouette gives 0.023 mm scatter in the recovered centre, whereas only a 40° arc — a partially occluded or partially detected boundary — degrades this to 1.00 mm mean and 4.06 mm maximum, because the cone axis becomes nearly degenerate with the in-plane direction. *Ray parallelism* governs the triangulation: the normal matrix $\sum_k (\mathbb{I} - \ell_k \ell_k^\top)$ is singular exactly when all rays are parallel, and near-parallel bundles reach condition numbers of order 10^6 without any warning from the linear solver. Since the emitter sits only 12.08 mm from the lens, the reflected rays can be uncomfortably close to parallel. Both the eigenvalue ratio of the silhouette fit and the condition number and residual of Eq. (24) should be reported for the real capture.

C. Principal direction and intensity

With $\mathbf{x}_s$ and μ known, the remaining parameters are recovered from images of a matte white plane. The nonlinearity in $\cos^\mu$ is removed by the change of variable

$$\mathbf{m}_s = \Psi^{1/\mu} \mathbf{n}_s, \quad (25)$$

which renders the model linear in $\mathbf{m}_s$ and solvable by least squares; the parameters are then recovered as

$$\Psi = \|\mathbf{m}_s\|^\mu, \quad \mathbf{n}_s = \mathbf{m}_s / \|\mathbf{m}_s\|. \quad (26)$$

Intensity is recovered up to one factor common to all emitters, which is absorbed into albedo.

Two qualifications are owed to a reviewer. First, the fit is least squares on the *transformed* response, not on intensity: since $\delta z/z = \mu^{-1} \delta I/I$, the transform rescales relative noise by $1/\mu$ and imposes an implicit per-pixel

weighting. Direction error accordingly scales as $1/\mu$, and the intensity estimate carries a small negative bias (measured at -0.009% under 1% multiplicative noise). The estimator is not best linear unbiased; it is a linearized least squares on a nonlinearly transformed response, and should be described as such. Second, the implementation clamps the Lambertian term at a small positive floor rather than discarding non-positive pixels, which can inject extreme outlier rows from shadowed or back-facing samples; with a flipped surface-normal convention it returns a plausible-looking direction with a catastrophically wrong intensity and no diagnostic. Shadowed pixels should be dropped and the surviving fraction reported (Appendix A, item C6).

Calibration captures do not exist

The near-light path is implemented and passes synthetic self-tests (Sec. VIII-F) but has *never been run on real data*, because the required captures have not been made: a mirror ball at ~ 10 poses per emitter and a matte white card at 3–5 tilts per emitter. The solver deliberately *requires* a solved calibration file rather than falling back on assumed ring symmetry — an assumed-geometry fallback is precisely how the far-field result came to look credible. This is the single blocking item for a physically correct depth channel. See Appendix A, item C1.

VI. RETROREFLECTIVE FIDUCIAL LOCALIZATION

Fiducials are retroreflective glass-bead disks affixed to the specimen. They serve two roles: they are the correspondences for registration (Sec. VII), and they are the points at which the depth channel is checked against CT (Sec. VIII-C).

A. The sparkle cue

Detection does not threshold brightness, which would confuse fiducials with specular glints on wet bone. The discriminating cue is texture: the glass-bead coating produces high-frequency sparkle where bone is smooth. A local high-frequency energy map is formed as the local RMS of the Laplacian,

$$T = \sqrt{G_\sigma * (\nabla^2 I)^2}, \quad \sigma = 3 \text{ px}. \quad (27)$$

B. Candidates and acceptance

Two detectors are pooled: a blob detector on the thresholded sparkle map, retaining components whose area, bounding-box aspect ratio (> 0.45) and fill ratio (> 0.45) are consistent with a solid disk; and a Hough circle detector over radii 12–60 pixel. Each candidate is scored over an inner disk of radius $0.8r$ and an annulus from $1.1r$ to $1.7r$, yielding interior sparkle, interior-minus-ring contrast, and raw brightnesses. Overlapping candidates within 48 pixel are resolved in favour of the sparkliest.

Acceptance requires clearing a raw-brightness floor and then satisfying any one of three signatures (Table IV). The ring-brightness ceiling in the **bright** path is what separates

TABLE IV
FIDUCIAL ACCEPTANCE SIGNATURES.

Signature	Condition	Rationale
strong	sparkle ≥ 130	unambiguous bead texture
ringed	sparkle ≥ 80 , contrast ≥ 25	dim bead, dark rim confirms
bright	raw ≥ 130 , contrast ≥ 30 , ring raw < 145	saturated bead, sparkle washed out

a fiducial from a glint: a genuine retroreflector sits in a dark surround, whereas a glint on bright bone is surrounded by bright bone and is rejected.

C. Cross-frame consolidation

Because the camera and fiducials are static while only the illumination changes (Table I), a physical fiducial appears at essentially the same pixel in every frame that captured it. Detections are clustered across the four single-emitter frames, their positions averaged, and the number of frames in which each was independently detected is recorded as n_{frames} .

This is the pipeline’s per-fiducial confidence statistic. A fiducial at $n_{\text{frames}} = 4$ has been confirmed four times under four different illumination directions; $n_{\text{frames}} = 1$ warrants visual review. It costs nothing — the frames are captured anyway for the photometric solve — and it is reported alongside every coordinate.

D. Fiducial coordinates

Depth at a fiducial is read as the *median* over its disk, for robustness to residual specular contamination. The reported triple is

$$(x_k, y_k, z_k) = (u_k s, v_k s, \text{median}_{\mathcal{D}_k} z). \quad (28)$$

Note the asymmetry: x and y are true in-plane millimetres, while z from the far-field baseline is a *relative* depth about the session median. Only the first two are metric in the ordinary sense — a distinction that becomes important in Sec. VIII-C.

Because the specular-removal step inpaints the strongly non-Lambertian bead from its surroundings, the depth sampled at a fiducial is that of the *bone seat beneath it*, not of the bead itself. This is the intended behaviour.

VII. REGISTRATION TO CT

PRINCIPAL SECTION — NOT YET WRITTEN

Sections VII-C–VII-E are scaffolding. No registration method has been finalized and no registration result exists. Every bracketed value is a placeholder, not a measurement. Do not circulate. See Appendix A, items R1–R8.

TABLE V
CT GROUND-CONTROL POINTS FOR THE ONE SESSION WITH CT (SHOT_011), IN THE CT FRAME. FIDUCIAL 2 HAS NO CT CORRESPONDENCE.

Fiducial	X (mm)	Y (mm)	Z (mm)
1	9.224	−1.632	11.818
3	3.729	−13.209	23.259
4	−17.665	0.732	17.412

A. Error nomenclature

Three error quantities are routinely conflated in this literature, and the distinction determines what may be claimed. *Fiducial localization error* (FLE) is the error in locating a fiducial in each modality. *Fiducial registration error* (FRE) is the residual at the points used to fit the transform. *Target registration error* (TRE) is the error at points *not* used to fit it.

This paper will report **TRE as its accuracy metric**. FRE will be reported only as a within-registration diagnostic, accompanied by the explicit statement that FRE is uncorrelated with TRE and therefore carries no information about registration accuracy for a given case [3], [4]. Leave-one-out surrogates are likewise not offered as a substitute, having also been reported as poor indicators of true TRE [5]. Evaluation targets must be points not used to compute the transform; if any evaluation target doubles as a registration fiducial, that must be disclosed, since the resulting TRE is optimistically biased.

B. Transfer into the camera frame by P3P

CT coordinates live in the scanner frame, so they cannot be compared with camera depths until a pose is solved. For the one session with CT ground truth, three fiducials have known CT positions (Table V). The pipeline solves the perspective-three-point problem on these correspondences using the measured intrinsics of Eq. (6), via OpenCV’s SOLVEPNP_AP3P, which implements the algebraic solution of Ke and Roumeliotis [2]. Points are then transferred by

$$\mathbf{x}_{\text{cam}} = \mathbf{R} \mathbf{x}_{\text{CT}} + \mathbf{t}, \quad (29)$$

and the lens-to-fiducial distance read as $\|\mathbf{x}_{\text{cam}}\|$, the camera centre being the frame origin.

Resolving the P3P ambiguity. Three correspondences admit up to four geometrically valid poses. Rather than trust the solver’s ordering, the pipeline enumerates the candidates and applies two filters: *cheirality*, requiring all points in front of the camera ($Z > 0$); and a *physical ordering* constraint, requiring the pose to place fiducial 4 nearest the lens, which was independently confirmed about the capture. If no pose satisfies both, the pipeline raises rather than returning a best-effort answer.

We characterised this strategy rather than assuming it works. Over 23365 random valid three-point configurations, the solver returned two poses in 92.6 % of cases and four in 7.4 %. Two findings follow. The cheirality filter is effectively a *no-op*: the P3P formulation already solves for positive

depths along the rays, so every returned pose passed it. And the ordinal constraint is a one-bit test against up to four candidates, so it is not sufficient in general — in 2.9% of those configurations at least two poses survived both filters. With only three correspondences the reprojection residual is at machine precision for *every* solution and cannot break the tie; the standard remedy is a fourth correspondence.

For the dataset here the disambiguation is nonetheless unambiguous: the solver returns two poses, both cheirality-valid, and exactly one places fiducial 4 nearest. That remained true in every trial of 20000 repetitions under 2 pixel centroid noise and 20000 more sweeping f_x over 400–900 pixel. We therefore report the technique as sound *for this configuration*, with the 2.9% general failure mode stated.

Latent implementation divergence

Two scripts implement this same selection with different tie-break rules: one returns the first surviving pose, the other assigns without breaking the loop and so retains the last. On this dataset exactly one pose survives, so no published number is affected, but the divergence would bite in the ambiguous regime above. Both should raise when more than one candidate survives — which would also answer the pose-count reporting item (Appendix A, R7).

C. Transform model

The registration problem is to recover the similarity transform $\mathcal{T} = (\sigma, R, \mathbf{t})$ carrying endoscopic measurements into the CT frame, the uniform scale σ being required because the photometric depth channel is relative rather than metric:

$$\mathcal{T}^* = \arg \min_{\sigma, R, \mathbf{t}} \sum_{k=1}^n \left\| \sigma R \mathbf{x}_k^{\text{endo}} + \mathbf{t} - \mathbf{x}_k^{\text{CT}} \right\|^2. \quad (30)$$

[METHOD TO BE WRITTEN.] The intended approach is fiducial-based: the retroreflective markers of Sec. VI are also identifiable in the CT volume, giving direct correspondences and hence a closed-form solution rather than an iterative surface fit. The estimator must be named precisely (closed-form point-set versus iterative; weighted or unweighted), the guard against the reflection failure mode reported ($\det R = +1$ verified on every registration), and error decomposed by parameter — rotation, translation and scale separately — rather than collapsed into one lumped residual.

[UNRESOLVED:] the relationship between the P3P pose of Sec. VII-B and this section must be settled. The P3P solution is already a rigid registration of three fiducials, and may constitute the paper’s existing registration result; whether the final method generalizes or replaces it is undecided.

D. Fiducial configuration

[REQUIRED, NOT OPTIONAL.] Configuration shape governs TRE: widely spread fiducials with the

configuration centroid near the target minimize it, while near-collinear configurations far from the target maximize it [6]. This subsection must report, per session, the number of fiducials, their coordinates, their spatial spread, the centroid position relative to each target, and any near-collinearity or near-coplanarity. The configuration must be justified, not merely counted.

The FLE model must also be stated rather than assumed: whether FLE is modelled as zero-mean isotropic Gaussian or as anisotropic, and how it was estimated. Every TRE prediction is a function of FLE, and the two modalities here plausibly differ — CT voxel anisotropy on one side, sub-pixel centroid precision on the other.

E. Validation protocol

[TO BE WRITTEN.] A five-part skeleton is intended: design of the evaluation dataset; ground-truth definition *and its own accuracy*; evaluation criteria; evaluation metrics; evaluation protocol. Accuracy, precision, robustness and stability are to be reported as four separate quantities.

Leave-one-fiducial-out cross-validation is the obvious protocol, but with only three CT-known fiducials in a single session it is not currently executable — a data-collection dependency, not an analysis one. The number of independent registrations per condition and how repeats were generated must be stated: a TRE from a single registration is an anecdote, not a measurement.

Ground-truth uncertainty not yet quantified

A validation cannot resolve error below the uncertainty of its own reference standard. The CT voxel size, reconstruction kernel, and the localization uncertainty of fiducial centroids *within* the CT volume are not recorded, so no accuracy claim finer than that unstated floor is defensible. Appendix A, item R8.

F. A cautionary prior result

An earlier exploratory attempt in this project registered dense reconstructed surface patches to the CT bone surface by automatic similarity ICP *without* correspondences. It produced a superficially attractive median surface deviation near 1.9 mm that was, on inspection, an artifact: the patches draped onto a single region of the bone rather than localizing to their true anatomical sites, and a near-planar patch resting on a curved surface yields a low residual *wherever* it is placed. The tell was uniformity — a standard deviation of only 0.17 mm across twelve patches at supposedly twelve different sites. That approach is not used here and no surface-ICP accuracy figure is reported. Correspondence-free ICP of small near-planar patches onto large curved surfaces is under-constrained in a way that does not announce itself in the residual.

VIII. EXPERIMENTAL VALIDATION

A. Dataset

Five capture sessions (shot_011–shot_015) of an ex-vivo vertebral specimen were processed: 5 sessions $\times$ 5

TABLE VI
FIDUCIAL DETECTION AND FAR-FIELD RELATIVE DEPTH BY SESSION.
SOURCE: `MARKER_DEPTHS.CSV`.

Session	N	z range (mm)	span	radius (px)	n_{frames}
shot_011	4	−2.302 ... +3.011	5.31	17.7–26.8	4,4,4,3
shot_012	3	−3.389 ... +2.295	5.68	25.8–26.5	4,4,4
shot_013	3	−2.600 ... +1.673	4.27	22.0–26.8	4,4,4
shot_014	4	−5.087 ... +0.197	5.28	12.0–29.3	1,4,4,3
shot_015	4	−1.168 ... +2.522	3.69	18.0–31.8	4,4,4,1
Pooled	18	−5.087 ... +3.011	—	12.0–31.8	78 % at 4

exposures = 25 images, yielding 18 consolidated fiducials. CT ground truth exists for one session and three of its four fiducials. This is a small n , and it is bounded by a stated constraint rather than by convenience: expanding the check requires additional CT acquisition, not additional endoscopy.

Specimen description missing

Species, anatomy, preparation, fiducial attachment method and physical fiducial diameter are not recorded. The physical diameter would independently validate the field-width scale of Eq. (5) against the measured 12.0–31.8 pixel radii. Appendix A, item M3.

B. Fiducial detection yield

Detection recovered **18 fiducials** (Table VI). Of these, **14 (78 %) were independently detected in all four single-emitter frames**. Two were seen in three frames, and two — one each in `shot_014` and `shot_015` — in only one, and are treated as provisional. Radii averaged 24.6 pixel (range 12.0–31.8 pixel), i.e. roughly 0.7–1.8 mm at the scale of Eq. (5). The two single-frame detections are among the smallest, consistent with confidence degrading at the small-radius end.

Pooled across sessions the far-field relative depth has mean −0.047 mm and standard deviation 2.25 mm, with per-session spans of 3.69–5.68 mm. These magnitudes are physically plausible for vertebral relief over a 37 mm field — which is exactly why the failure reported next is not self-announcing.

C. Far-field depth against CT: the inversion

This is the paper’s principal empirical result.

For the three CT-known fiducials, the P3P solution of Sec. VII-B — the branch placing fiducial 4 nearest — gives lens distances of 33.5 mm (fiducial 4), 50.9 mm (1) and 53.2 mm (3), spanning 19.7 mm. Converted to camera-frame depth relative to the fiducial-set median, the CT values span 20.7 mm. The far-field photometric depths for the same three fiducials are +0.087, −2.302 and +2.673 mm (Table VII).

Three findings follow (Fig. 3).

(i) **The correlation is negative.** Pearson correlation between far-field relative depth and CT lens-to-fiducial distance is $r = -0.923$; against camera-frame CT depth

TABLE VII
FAR-FIELD PHOTOMETRIC DEPTH VERSUS CT GROUND TRUTH,
`SHOT_011`. ALL VALUES IN MILLIMETRES.

Fid.	z_{est}	CT z_{rel}	CT dist.	z_{corr}	resid.
1	+0.087	0.000	50.9	−6.157	−6.157
3	−2.302	+0.719	53.2	+3.920	+3.201
4	+2.673	−20.023	33.5	−17.066	+2.957
2	+3.011	—	—	−18.492	—

it is $r = -0.891$. The depth channel varies inversely with true distance.

(ii) **The ordering is reversed.** Ranked nearest to farthest, photometry gives (3, 1, 4) where CT gives (4, 1, 3). This is not noise degrading a correct ordering; it is a systematic sign inversion.

(iii) **Affine correction cannot rescue it.** Least-squares affine fitting onto the CT control points returns

$$z_{\text{corr}} = -4.218 z_{\text{est}} - 5.791 \text{ mm}, \quad (31)$$

whose *negative slope* states the inversion algebraically. Even permitting the sign flip, the fit leaves 4.36 mm RMS across a 20.7 mm true span — about 21 % of the measured range. With two free parameters and three control points, this residual reflects genuine lack of fit rather than an interpolation artifact.

The corrected map is retained as a diagnostic only: an affine rescale with slope −4.218 amplifies the underlying far-field spatial bias roughly threefold while merely re-seating the fiducial values.

This is the empirical case for near-light modelling. The failure is a predictable consequence of $\eta = 0.40$ (Sec. IV-A), not of implementation, and it cannot be repaired downstream: the artifact is spatially varying, so a global affine correction cannot remove it, and the tilt-plane alternative needs three non-collinear control fiducials, which only one session provides — and there it interpolates exactly, yielding no residual information. The forward model must be fixed instead.

D. Lens-to-fiducial distances

Eighteen distances were computed, of which three are CT-validated and fifteen photometric and flagged unreliable. The CT-validated values are 50.9, 53.2 and 33.5 mm; these are trustworthy up to the $\sim 1.5\times$ scale ambiguity of Sec. III-C, which is multiplicative and therefore affects neither their ordering nor their ratios. The photometric values have mean 48.6 mm and standard deviation 4.4 mm.

The apparent tightness of that distribution is *not* evidence of accuracy. Those values are constructed by adding a small relative depth (SD 2.25 mm) to a *constant* 46.0 mm anchor and back-projecting, so their spread is largely inherited from the anchor by construction — and the component that does vary is the very quantity shown above to be sign-inverted.

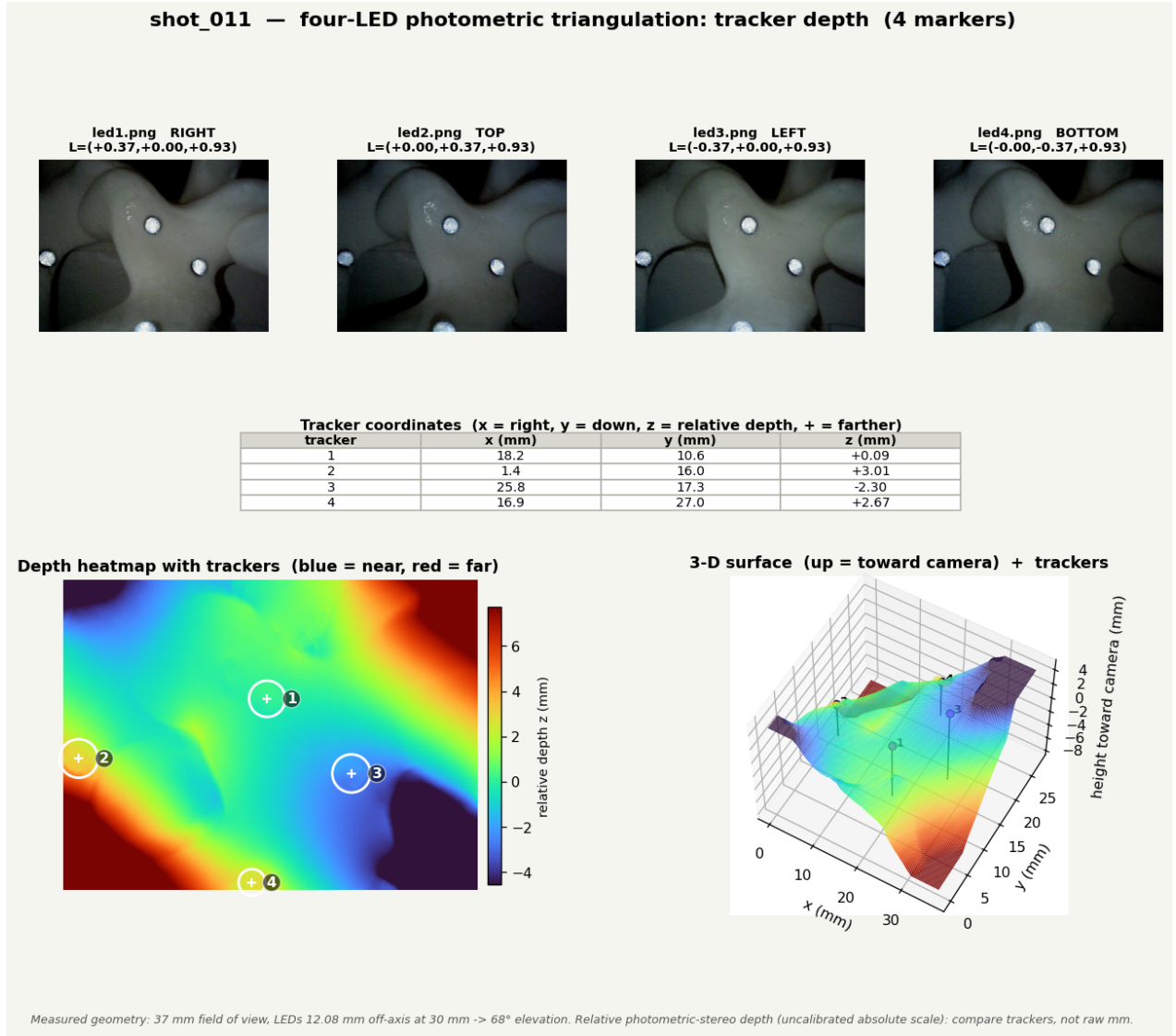


Fig. 2. Per-session composite output for **shot_011**. Top row: the four single-emitter exposures with their light vectors. Middle: recovered fiducial coordinates. Bottom left: the relative depth field with all four detected fiducials circled and numbered. Bottom right: the same field as a 3-D surface with a vertical stem at each fiducial. The depth field shown is the *far-field* baseline subsequently shown to be inverted (Sec. VIII-C); it is smooth and anatomically plausible, and nothing in it signals the failure.

Two defects in the anchor itself

The implementation documents 46.0 mm as the *median* P3P distance. It is not: the three CT-validated distances have median 50.92 mm and mean 45.84 mm, so the constant matches the mean, not the median. Its provenance is genuinely ambiguous — 45.9 mm is also the working distance implied by the checkerboard focal length — and it must be settled and the label corrected, since fifteen of the published rows depend on it. Separately, the anchor is a *Euclidean* distance statistic but is used as an *axial* depth Z before multiplication by the ray norm, applying the obliquity factor twice: negligible (0.59%) for a near-axial fiducial but 15.3% for the most off-axis one. Both are small against the declared unreliability of these rows, but both are definite errors.

E. A scale conflict no focal length resolves

Section III-C reported the focal-length discrepancy as a $\sim 1.53\times$ ambiguity. Cross-checking the pose solution against the bench geometry reveals something stronger, and we report it because a reviewer holding Table V and the fiducial pixel coordinates can reproduce it in a few lines.

Feeding $f_x = 519$ pixel — itself *derived* from the premise $d_{\text{work}} = 30$ mm — into the P3P solve returns a median axial depth of 50.62 mm, 1.69 times that premise. This is not a focal-length ambiguity, because the P3P scale is fixed by the known metric size of the CT triangle, which makes the *implied* field width $\Omega' = WZ/f_x$ almost independent of f_x (Table VIII).

Across f_x from 300–2000 pixel the implied field width stays within 59.1–62.8 mm, i.e. 1.60–1.70 times the 37.0 mm bench figure, and a root search finds *no* f_x in 250–1400 pixel at which $\Omega' = 37$ mm. Neither candidate focal length

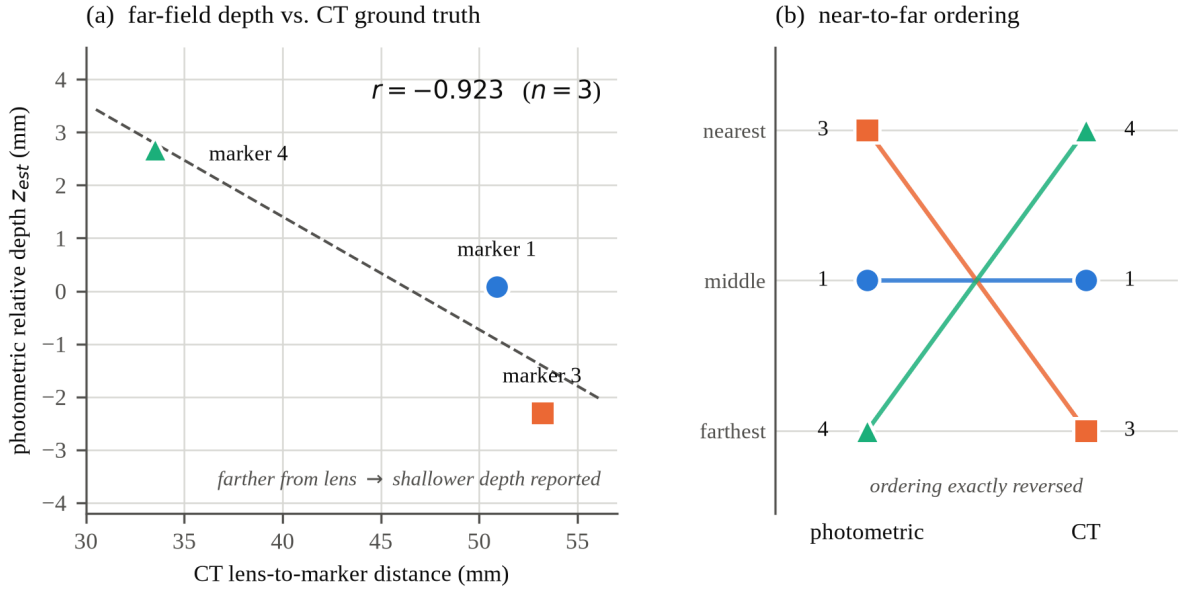


Fig. 3. The depth inversion at `shot_011`. (a) Far-field photometric relative depth against CT-derived lens-to-fiducial distance, with the least-squares trend (dashed). The relationship is negative: fiducials that CT places *farther* from the lens are assigned *shallower* depth. (b) Near-to-far ordering under each method. Fiducial 1 holds the middle position while 3 and 4 exchange the extremes, reversing the sequence (3, 1, 4) to (4, 1, 3). Identity is encoded by colour *and* symbol so the figure survives greyscale reproduction.

TABLE VIII
IMPLIED FIELD WIDTH AS A FUNCTION OF ASSUMED FOCAL LENGTH.
THE RECOVERED DEPTH SCALES WITH f_x , SO THE IMPLIED FIELD
WIDTH IS NEARLY INVARIANT — AND NEVER APPROACHES THE
MEASURED 37 MM.

f_x (px, at $W=640$)	300	519	794	1000	2000
median Z (mm)	28.6	50.6	75.3	93.9	184.6
implied Ω' (mm)	61.1	62.4	60.7	60.1	59.1

reconciles the two. The conflict is with the field-width measurement, not with f_x .

The same tension appears independently in the fiducial geometry. The CT pairwise separations are 17.18, 27.57 and 26.20 mm; at the 37 mm scale their ratios to the projected separations are 1.67–1.99, which requires the fiducial plane to be tilted 53.8° from the image plane. At the P3P-implied scale the ratios fall to 1.09–1.30, a far more natural configuration.

Unresolved — two candidate explanations, and we cannot yet distinguish them

(i) **The 37 mm field-width measurement is wrong** by a factor near 1.7. (ii) **The CT-to-detection correspondence is wrong.** The detector assigns fiducial indices purely by image position (top-to-bottom, then left-to-right); that ordering bears no necessary relation to the CT numbering, yet the distance code assumes detection index i equals CT fiducial i . This assumption is nowhere verified.

Enumerating all 24 assignments of the three CT points to the four detections, and keeping only those that place

fiducial 4 nearest with all distances physically possible, leaves eight candidates whose implied field widths are 44.2 mm, 44.4 mm, 45.5 mm, 46.8 mm, 47.0 mm, 47.1 mm, 58.7 mm and 62.4 mm. The pipeline’s own assignment is the *worst* of the eight, and **none reaches 37 mm**. Permutations that would imply ~ 37 mm each require either a fiducial at 6.7–7.8 mm from the lens (impossible) or fiducial 4 to be farthest, contradicting the confirmed physical ordering.

Required: re-measure the field width, and independently establish the CT-to-detection correspondence — then report the check either way. Until this is settled, absolute distances should be quoted with the conflict stated, not merely with the $\sim 1.53\times$ focal ambiguity. See Appendix A, items C2 and C8.

Note that this conflict does *not* weaken the inversion result of Sec. VIII-C: correlation and rank ordering are invariant under any positive rescaling of the CT distances, and the ordering constraint used to select the pose is unaffected.

F. Calibration self-tests

Both auxiliary modules ship self-tests on synthetic ground truth, runnable without captured data. Both pass (Table IX).

The passing affine- z test does *not* license the affine correction of Sec. VIII-C: it confirms the estimator recovers a scale-and-offset distortion, whereas the real distortion is a spatially varying near-light bias that this model is not designed to remove.

TABLE IX
SELF-TEST RESULTS ON SYNTHETIC GROUND TRUTH.

Module	Stage	Result
correction	affine-z, 2 control pts	0.015, 0.006, 0.008 mm
correction	tilt-plane, 3 control pts	0.030, 0.046 mm
LED calib.	anisotropy, $\theta_{1/2} = 60^\circ$	$\mu = 1.0000$
LED calib.	position (mirror ball)	0.00 mm
LED calib.	direction / intensity	0.090° / 0.01 %

More importantly, we must report what these tests do *not* establish, because we tested that question directly by mutation rather than by inspection.

The self-tests are not sound validation as currently written

Each test renders a synthetic scene with the same forward model the solver inverts, and generator and solver share the physical relations under test. We verified the consequences by deliberately corrupting the code:

- **The emitter-position test cannot distinguish correct from incorrect geometry.** Replacing $\sin\beta$ with $\tan\beta$ in the solver and $\arcsin$ with $\arctan$ in the generator — a fundamentally wrong tangent-cone relation — still passes, at 3×10^{-12} mm. Worse, running the *wrong* solver against a *correct* generator passes the shipped 2.0 mm threshold on 10 of 12 seeds, including the seed actually shipped. The true noise-free round-trip error is 2.5×10^{-12} mm, twelve orders of magnitude below the assertion: the threshold is not a tolerance but a blindfold.
- **The $\cos^4$ vignetting law and the $1/r^3$ falloff are tautological.** Each appears in identical form in generator and solver and cancels exactly; changing the exponent in both simultaneously leaves the reported errors bit-identical.
- **The linearization is never exercised.** The direction/intensity test hard-codes $\mu = 1$, where $(\cdot)^{1/\mu}$ is the identity and the change of variable $\mathbf{m}_s = \Psi^{1/\mu} \mathbf{n}_s$ degenerates. (Independent sweeps at $\mu \in \{0.51, 1, 2, 4.82\}$ do confirm exact recovery, so no bug is hiding here — but the shipped test provides no evidence of it.)
- **Error propagation between stages is untested.** Stage 2 receives the exact $\mathbf{x}_s$, so it never sees Stage 1’s error. Injecting a realistic 0.5 mm error in $\mathbf{x}_s$ produces 2.09° mean direction error and 1.18 % intensity error — the dominant real-world term, and invisible to the test.

Required before submission: tighten the position tolerance to $\sim 1 \times 10^{-9}$ mm; generate silhouettes by solving the ray-sphere tangency condition, which is algebraically independent of $\sin\beta = R/\|\mathbf{C}\|$; add a noisy variant with a justified tolerance; exercise $\mu \neq 1$; and report Stage-1→Stage-2 propagation. See

Appendix A, item C6.

What the tests *do* establish is that the linear algebra correctly inverts the model as the authors understand it. That is worth reporting, but it is not validation of the model’s fidelity to the rig, and the paper must not claim otherwise.

G. Registration results

NO REGISTRATION RESULT EXISTS

Every number in this subsection is invented scaffolding, present only so the results section has the correct shape. Nothing here may be cited or circulated.

Registration was evaluated over **[N]** correspondences from **[M]** sessions. The closed-form similarity solution returned scale **[S.SSS]**, with rotation, translation and scale residuals reported separately. **TRE** at independent targets: mean **[T.TT]**, median **[T.TT]**, SD **[T.TT]**, maximum **[T.TT]**, 95 %ile **[T.TT]** mm, reported per target rather than pooled. **FRE** (diagnostic only): **[F.FF]** mm RMS — reported for completeness and *not* an accuracy claim.

Sensitivity to the focal-length ambiguity must be reported as a paired analysis at $f_x = 519$ and 794 pixel, since a $1.5\times$ intrinsic error propagates directly into recovered scale and translation.

IX. DISCUSSION

A. What the pipeline supports today

The clearest outcome is a sharp, measured separation between reliable and unreliable outputs. Fiducial localization is robust: 78 % of fiducials were confirmed under four independent illumination conditions, and the three-signature acceptance logic with its dark-surround requirement separates retroreflectors from glints on wet bone. The CT/P3P distances rest on independent ground truth and a physically disambiguated pose. Both are suitable registration inputs.

The far-field depth channel is not. It is sign-inverted against the only available ground truth, and every quantity derived from it inherits that inversion. These outputs are retained and labelled as unreliable in the data files themselves, so the failure is documented rather than hidden.

We stress that this separation was possible *only* because one session carried CT ground truth. With no ground truth, the depth maps of Fig. 2 are entirely plausible — smooth, with anatomically sensible relief of a few millimetres, and fiducials at sensible depths. Nothing in the output signals that the ordering is reversed. This is a general hazard of shading-based reconstruction and argues for including ground-truth control points in at least one session of any such study as routine design.

B. The corrective path and its cost

The near-light solver and its calibration constitute a complete, implemented and unit-validated path to physically correct depth. The remaining cost is not development

but *data acquisition* — a mirror ball and a matte white card, one session’s work. Separately, the $1.53\times$ focal-length discrepancy must be resolved before any absolute metric claim; it is settled by imaging an object of known size at a known distance.

C. Implications for registration

[EXPAND ONCE SEC. VII IS WRITTEN.]

The inversion constrains the registration design directly, and favourably. Because the depth channel is unusable, registration cannot depend on it — but the fiducial pixel coordinates and the CT/P3P geometry are sound, and correspondence-based registration on fiducials uses exactly those. A fiducial-correspondence formulation is therefore not merely convenient but the only one the current data supports, and it is the more rigorous choice regardless: it yields a closed-form solution and a directly measurable target error, rather than an iterative fit whose residual can be low for reasons unrelated to correctness.

X. LIMITATIONS

- 1) The far-field depth channel is sign-inverted ($r = -0.923$), invalidating the depth column for all sessions and the fifteen photometric distances.
- 2) Ground truth exists for one session and three fiducials ($n = 3$). The correlation and ordering findings are unambiguous at this n ; the affine residual is a weak estimate.
- 3) The ground truth’s own uncertainty is unquantified.
- 4) Absolute scale is unresolved. Beyond the $\sim 1.53\times$ focal-length ambiguity, the pose solution implies a field width of 59–63 mm for *any* focal length, conflicting with the 37 mm bench measurement (Sec. VIII-E); either that measurement or the CT-to-detection correspondence is wrong. The calibration reprojection residual is also unrecorded.
- 5) The principal point is set to the image centre rather than its calibrated value, a 14.5 pixel vertical discrepancy worth 0.1–0.7 mm on the reported distances.
- 6) The calibration self-tests share their forward model with the solvers and, for emitter position, carry a tolerance loose enough to pass demonstrably wrong geometry (Sec. VIII-F). They validate implementation, not model fidelity.
- 7) The near-light path has never been run on real data, pending calibration captures.
- 8) Capture conditions — ambient control, auto-exposure and white-balance settings, saturation — are undocumented.
- 9) Two fiducials rest on single-frame detections.
- 10) Raw captures are not committed, so the pipeline cannot be re-executed end-to-end from the repository alone.
- 11) Software versions are unpinned, which additionally makes one solver citation version-dependent (Appendix A, item M5).
- 12) **[REGISTRATION LIMITATIONS — PENDING]**

XI. CONCLUSION

We have presented a near-light photometric stereo pipeline for a four-LED ring endoscope: the ring geometry and its offset and scale calibration, a calibrated point-source forward model, a perspective log-depth solver, per-emitter photometric calibration, retroreflective fiducial localization, and fiducial-anchored transfer into the CT frame.

Measured against CT, the far-field approximation fails at this geometry not by a scale factor but in sign: the recovered depth is anti-correlated with truth ($r = -0.923$) and reverses the ordering of all three CT-known fiducials, and no affine correction recovers it. With an offset-to-working-distance ratio of 0.40, this is the predictable consequence of applying a distant-source model to a near-light rig. The calibrated near-light formulation is the corrective path; it is implemented and passes synthetic validation, and is blocked only on one session of calibration captures.

Fiducial localization and CT/P3P distances are reliable and constitute the registration inputs the pipeline can support today. **[FINAL REGISTRATION CONCLUSION — PENDING]**

APPENDIX A OPEN ITEMS

Registration (principal claim). **R1** Decide and document the registration formulation; resolve its relationship to the existing P3P pose. **R2** Acquire per-session CT fiducial coordinates for `shot_012–015`. **R3** Produce the result: scale, per-parameter residuals, TRE per target, FRE as diagnostic. **R4** Define and execute a validation protocol with independent targets; state repetition count. **R5** Write the fiducial-configuration subsection and justify the configuration. **R6** Report sensitivity to the focal-length ambiguity at both f_x values. **R7** Log and report the P3P candidate-pose count and filter survival. **R8** Quantify CT ground-truth uncertainty as the validation noise floor.

Measurement and calibration. **C1** Capture LED calibration targets (mirror ball ~ 10 poses per emitter; matte white card 3–5 tilts); record $\theta_{1/2}$, the Arduino Pro variant, LED part number and drive current. **C2** Resolve the f_x ambiguity; record reprojection residual, pose count and distortion coefficients. **C3** Run the near-light solver on real data and re-validate against CT. **C4** Re-derive lens-to-fiducial distances from near-light depth. **C5** Obtain CT ground truth for the remaining fiducial in `shot_011`. **C6** Repair the calibration self-tests: tighten the position tolerance to $\sim 1 \times 10^{-9}$ mm, generate silhouettes by tangency solving rather than by the cone parameterisation the solver inverts, add a noisy variant, exercise $\mu \neq 1$, and report inter-stage error propagation. Drop shadowed pixels in the direction/intensity fit instead of clamping, and report the surviving fraction. Emit conditioning diagnostics for the silhouette fit and the ray triangulation. **C7** Record the measured E_R/E_B of the emitters on a neutral target, and state the datasheet angle convention ($\theta_{1/2}$ versus $2\theta_{1/2}$) alongside the recorded value. **C8** Resolve the field-width conflict of Sec. VIII-E: re-measure the field width, and independently establish the CT-to-detection fiducial correspondence rather than assuming index equality. **C9** Correct the intrinsics: use the calibrated principal point rather than the image centre, and fix the anchor-constant label and the axial/Euclidean double-count in the photometric distance path.

Manuscript. **M1** Authors, affiliations, funding, competing interests, ethics and specimen provenance (naming the oversight or licensing body), consent, data and code availability, author contributions, AI-tool disclosure. **M2** Verify every reference against the publisher record — several entries are currently unverified (Appendix B). **M3** Specimen description and fiducial physical diameter. **M4** Figure preparation at journal resolution. **M5** Pin software versions — required for correctness of the solver citation, not only reproducibility. **M6** Select target journal and reformat; abstract structure, declaration headings and word budgets differ, and several budgets are inclusive of references and captions.

APPENDIX B

REFERENCE VERIFICATION STATUS

Reference metadata is a common source of published error, and entries drawn from third-party bibliography files are frequently corrupt. Publisher sites were not reachable from the build environment, so most entries below could not be confirmed against a publisher record.

Verified: [1] (confirmed against the corresponding author’s released code repository); [2] (confirmed as the algorithm implemented by OpenCV `SOLVEPNP_AP3P`, by direct inspection of the OpenCV source and its bibliography; proceedings pagination unverified).

Unverified — must be checked before submission: [3], [4], [5], [6], [7], [8].

Version-dependent citation. The neighbouring OpenCV flag `SOLVEPNP_P3P` maps to Gao *et al.* (2003) in OpenCV 4.5.5 but to a different algorithm on current 4.x releases. Our code calls `SOLVEPNP_AP3P`; pin the version or the citation is unfalsifiable.

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